הודיה חנניאייב - 324857549

תהילה חננאייב – 325435956

מערכות הפעלה מעבדה – תרגיל 3

שלב 1:

```
tehila2003@DESKTOP-OPO0EU0: ~
tehila2003@DESKTOP-OPO0EU0:~$  gdb --args hello
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from hello...
(gdb) b main
Breakpoint 1 at 0x11b8: file hello.c, line 3.
```

```
tehila2003@DESKTOP-OPO0EU0: ~
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from hello...
(gdb) b main
Breakpoint 1 at 0x11b8: file hello.c, line 3.
(gdb) run
Starting program: /home/tehila2003/hello
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main (argc=1, argv=0xffffd274) at hello.c:3
3           printf("hello %s\n", argv[1]);
(gdb) print &"hello %s\n"
$1 = (char (*)[10]) 0x5655a1a0
(gdb) print argv[1]
$2 = 0x0
(gdb)
```

שלב 2:

```
tehila2003@DESKTOP-OPO0EU0:~$ gcc -m32 -g fork.c -o fork
tehila2003@DESKTOP-OPO0EU0:~$ ./fork&
[2] 48316
tehila2003@DESKTOP-OPO0EU0:~$
 My identifier is ID = [48316]

 My identifier is ID = [48317]
^C
tehila2003@DESKTOP-OPO0EU0:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 10:55 ?        00:03:56 /sbin/init
root           2       1  0 10:55 ?        00:00:00 /init
root           6       2  0 10:55 ?        00:00:00 plan9 --control-socket 6 --
root          38       1  0 10:55 ?        00:00:00 /lib/systemd/systemd-journa
root          58       1  0 10:55 ?        00:00:01 /lib/systemd/systemd-udevd
```

ניתן לראות את שני התהליכים לפני הריגתם (מסומנים בצהוב) :

```
tehila2003@DESKTOP-OPO0EU0: ~
root         435     330  0 10:55 ?        00:01:30 python3 /snap/ubuntu-deskto
root        3139       1  0 11:05 ?        00:00:00 /usr/libexec/packagekitd
root        3144       1  0 11:05 ?        00:00:00 /usr/libexec/polkitd --no-d
root        8086       2  0 11:25 ?        00:00:00 /init
root        8087    8086  0 11:25 ?        00:00:00 /init
tehila2+    8094    8087  0 11:25 pts/0    00:00:00 -bash
tehila2+   48175    8094 99 14:16 pts/0    00:00:54 ./fork
tehila2+   48176   48175 99 14:16 pts/0    00:00:54 ./fork
tehila2+   48316    8094 99 14:17 pts/0    00:00:20 ./fork
tehila2+   48317   48316 99 14:17 pts/0    00:00:20 ./fork
tehila2+   48398    8094  0 14:17 pts/0    00:00:00 ps -ef
tehila2003@DESKTOP-OPO0EU0:~$ kill -9 48316
tehila2003@DESKTOP-OPO0EU0:~$ kill -9 48317
[2]+  Killed                  ./fork
tehila2003@DESKTOP-OPO0EU0:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 10:55 ?        00:03:57 /sbin/init
root           2       1  0 10:55 ?        00:00:00 /init
root           6       2  0 10:55 ?        00:00:00 plan9 --control-socket 6 --
```

בתמונה זו אפשר לראות שהתהליכים נהרגו כנדרש.

```
tehila2003@DESKTOP-OPO0EU0: ~
root         214       1  0 10:55 hvc0     00:00:00 /sbin/agetty -o -p -- \u --
root         217       1  0 10:55 tty1     00:00:00 /sbin/agetty -o -p -- \u --
root         330     208  0 10:55 ?        00:00:16 /snap/ubuntu-desktop-instal
root         359       2  0 10:55 pts/1    00:00:00 /bin/login -f
tehila2+     417       1  0 10:55 ?        00:00:00 /lib/systemd/systemd --user
tehila2+     418     417  0 10:55 ?        00:00:00 (sd-pam)
tehila2+     423     359  0 10:55 pts/1    00:00:00 -bash
root         435     330  0 10:55 ?        00:01:31 python3 /snap/ubuntu-deskto
root        3139       1  0 11:05 ?        00:00:00 /usr/libexec/packagekitd
root        3144       1  0 11:05 ?        00:00:00 /usr/libexec/polkitd --no-d
root        8086       2  0 11:25 ?        00:00:00 /init
root        8087    8086  0 11:25 ?        00:00:00 /init
tehila2+    8094    8087  0 11:25 pts/0    00:00:00 -bash
tehila2+   48175    8094 99 14:16 pts/0    00:02:08 ./fork
tehila2+   48176   48175 99 14:16 pts/0    00:02:08 ./fork
tehila2+   48694    8094  0 14:18 pts/0    00:00:00 ps -ef
tehila2003@DESKTOP-OPO0EU0:~$
```

ביצענו שוב רק הפעם עם script כדי ללכוד את המסך:

(*הערה: מכיוון שביצענו שוב ה pid של התהליכים שונה)

```
tehila2003@DESKTOP-OPO0EU0: ~
  GNU nano 6.2                                        A
Script started on 2024-05-16 14:26:55+03:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="79" LINES="19"]
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$ gcc -m32 >
^[[?2004l^M^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m>
^[[?2004l^M[1] 50770
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$
 My identifier is ID = [50770]

 My identifier is ID = [50771]
^C^[[?2004l^M^[[?2004h^[[?2004l^M
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$ ps -ef
^[[?2004l^MUID         PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 10:55 ?        00:04:02 /sbin/init
root           2       1  0 10:55 ?        00:00:00 /init
root           6       2  0 10:55 ?        00:00:00 plan9 --control-socket 6 --
root          38       1  0 10:55 ?        00:00:00 /lib/systemd/systemd-journa
root          58       1  0 10:55 ?        00:00:01 /lib/systemd/systemd-udevd
root          68       1  0 10:55 ?        00:00:00 snapfuse /var/lib/snapd/sna
root          69       1  0 10:55 ?        00:00:01 snapfuse /var/lib/snapd/sna
root          72       1  0 10:55 ?        00:00:00 snapfuse /var/lib/snapd/sna
root          75       1  0 10:55 ?        00:00:00 snapfuse /var/lib/snapd/sna
root          81       1  0 10:55 ?        00:00:00 snapfuse /var/lib/snapd/sna
root          84       1  0 10:55 ?        00:00:03 snapfuse /var/lib/snapd/sna
root          90       1  0 10:55 ?        00:00:00 snapfuse /var/lib/snapd/sna
root          91       1  0 10:55 ?        00:00:01 snapfuse /var/lib/snapd/sna
systemd+     101       1  0 10:55 ?        00:00:00 /lib/systemd/systemd-resolv
root         122       1  0 10:55 ?        00:00:00 /usr/sbin/cron -f -P
message+     124       1  0 10:55 ?        00:00:00 @dbus-daemon --system --add
root         137       1  0 10:55 ?        00:00:00 /usr/bin/python3 /usr/bin/n
syslog       142       1  0 10:55 ?        00:00:00 /usr/sbin/rsyslogd -n -iNON
root         143       1  0 10:55 ?        00:00:03 /usr/lib/snapd/snapd
root         148       1  0 10:55 ?        00:00:00 /lib/systemd/systemd-logind

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

<u>שלב 3:</u>

```
tehila2003@DESKTOP-OPO0EU0:~$ script B
Script started, output log file is 'B'.
tehila2003@DESKTOP-OPO0EU0:~$  gcc -o fork1 fork1.c
tehila2003@DESKTOP-OPO0EU0:~$ ./fork1

 [63887] L0

 [63887] L1

 [63888] Bye

 [63887] L2

 [63889] Bye

 [63887] Bye

 [63890] Bye
tehila2003@DESKTOP-OPO0EU0:~$ exit
exit
Script done.
tehila2003@DESKTOP-OPO0EU0:~$ pico B
tehila2003@DESKTOP-OPO0EU0:~$
```

```
tehila2003@DESKTOP-OPO0EU0: ~
  GNU nano 6.2                                        B
Script started on 2024-05-16 15:22:49+03:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="115" LINES="35"]
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$  gcc -o f>
^[[?2004l^M^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m>
^[[?2004l^M
 [63887] L0

 [63887] L1

 [63888] Bye

 [63887] L2

 [63889] Bye

 [63887] Bye

 [63890] Bye
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$ exit
^[[?2004l^Mexit

Script done on 2024-05-16 15:23:16+03:00 [COMMAND_EXIT_CODE="0"]
```

שלב 4:

```
tehila2003@DESKTOP-OPO0EU0:~$ script C
Script started, output log file is 'C'.
tehila2003@DESKTOP-OPO0EU0:~$  gcc -o fork2 fork2.c
tehila2003@DESKTOP-OPO0EU0:~$ ./fork2

 [72770] L0

 [72770] Bye

 [72771] L1

 [72771] Bye

 [72772] L2
tehila2003@DESKTOP-OPO0EU0:~$
 [72772] Bye

 [72773] Bye
^C
tehila2003@DESKTOP-OPO0EU0:~$ exit
exit
Script done.
tehila2003@DESKTOP-OPO0EU0:~$ pico C
tehila2003@DESKTOP-OPO0EU0:~$
```

```
tehila2003@DESKTOP-OPO0EU0: ~
  GNU nano 6.2                                                    C
Script started on 2024-05-16 16:00:15+03:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="115" LINES="35"]
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$  gcc -o f>
^[[?2004l^M^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m>
^[[?2004l^M
 [72770] L0

 [72770] Bye

 [72771] L1

 [72771] Bye

 [72772] L2
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$
 [72772] Bye

 [72773] Bye
^C^[[?2004l^M^[[?2004h^[[?2004l^M
^[[?2004h^[]0;tehila2003@DESKTOP-OPO0EU0: ~^G^[[01;32mtehila2003@DESKTOP-OPO0EU0^[[00m:^[[01;34m~^[[00m$ E^H^[[Ke^>
^[[?2004l^Mexit

Script done on 2024-05-16 16:00:51+03:00 [COMMAND_EXIT_CODE="130"]
```